

CH 5440 Multivariate Data Analysis in Process Monitoring and Diagnosis

Assignment 5

(1) For the flow process shown in Fig. 1, flow rates of all streams are measured and given in file `flowdata.mat`. The errors in different flow measurements have different unknown error variances.

(a) Assuming the errors in different variables are independent, apply IPCA to estimate diagonal error variances and identify the linear steady state model relating the flow variables (assuming that the number of constraints is known). For this purpose use the function `stdest.m`, which uses a least squares procedure for estimating diagonal error variances. This function takes the constraint model and data as inputs and returns the standard deviation of errors as output.

(b) Assume that the number of constraints is unknown and estimate it using hypothesis test for equality of eigenvalues, along with the error standard deviations and model constraints.

(c) Assume that the errors in measurement of variables F_3 and F_4 are correlated. In this case, the error covariance matrix is not diagonal, but also contains an off-diagonal element corresponding to the element (3, 4) and (4,3) in the error covariance matrix. Modify the `stdest.m` function to estimate the diagonal and off-diagonal elements of the error covariance matrix and use it to extend the IPCA method to simultaneously estimate the number of constraints, elements of the error covariance matrix and constraint model

For each of the above cases, report the converged eigenvalues, error variances, and the maximum absolute difference between the elements of the true and estimated regression matrices corresponding to the choice of F_1 and F_2 as independent variables.

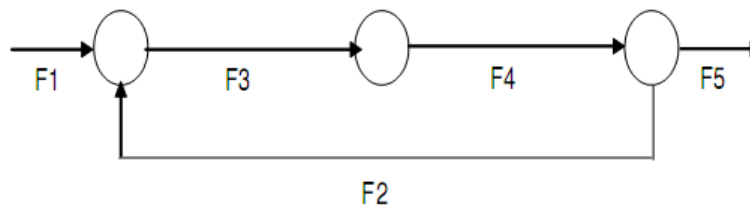


Fig. 1. Schematic of a flow process